SemanticFlow: Semantic Segmentation of Sequential LiDAR Point Clouds From Sparse Frame Annotations

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***Abstract*— Sequential point clouds acquired by light detection and ranging (LiDAR) technology provide accurate spatial information for environmental sensing. However, semantic segmentation of point cloud sequences relies on many manual point-wise annotations, which are error-prone and expensive. Existing mainstream weakly supervised methods tackle this by reducing the percentage of labeled points, but they are mostly designed for static indoor scenes and are hard to apply practically. From the viewpoint of realistic annotation procedures and the nature of point cloud sequences, this article proposes a novel semantic segmentation method, SemanticFlow, for LiDAR point cloud sequences using sparse frames with annotations. The proposed method achieves competitive performance compared with fully supervised methods. Specifically, we designed a bidirectional cross-frame pseudolabel propagation module that uses scene flow to learn the correlation and propagate pseudolabels across neighboring frames. In addition, a label refinement mechanism is proposed to select reliable pseudolabels for learning. Extensive experiments on SemanticKITTI, SemanticPOSS, and Synthia 4-D datasets demonstrate that our sparse frame annotation method is compatible with some fully supervised counterparts.**

***Index Terms*— Light detection and ranging (LiDAR) point cloud sequences, semantic segmentation, sparse frame annotation.**

I. Introduction

INTERPRETING the semantic composition from light detection and ranging (LiDAR) point clouds is crucial for autonomous driving and smart cities. However, this is particularly challenging as it requires accurately analyzing large-scale and dynamic point cloud sequences, in which a simple object may have vastly different partial shapes over the sequence due to visual occlusion and complex motion dynamics.

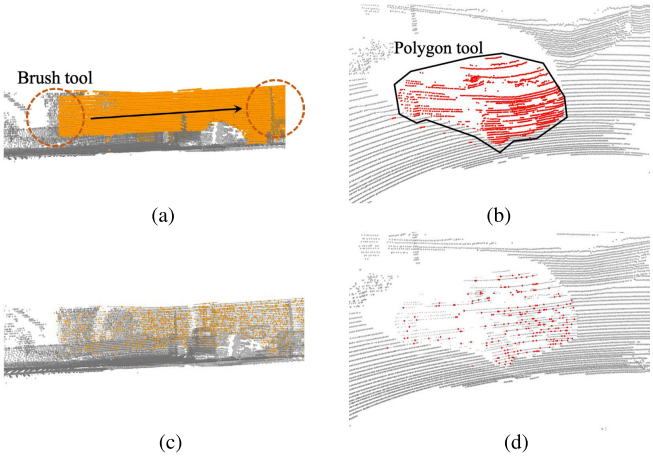


Fig. 1. Illustration of practical annotation tools and random point labeling. (a) Brush tool, (b) polygon tool, and (c) and (d) random labeling at 10%.

Inspired by the seminal works [1], [2], sophisticated techniques have been developed to learn semantics for point clouds. These methods can generally be divided into two categories: single scan pipelines [3], [4], [5], [6], and sequential scan-based methods [7], [8], [9], [10]. Although remarkable semantic segmentation accuracy has been achieved on single scans, the first category of methods is specifically designed to learn local spatial patterns, ignoring the motion patterns of objects across multiple scans or even longer sequences. This issue is solved by the second category of approaches designed to capture motion patterns between multiple frames, achieving encouraging results in 3-D semantic segmentation over sequences. However, these methods rely on large-scale and per-frame human annotations to train the networks.

However, due to the enormous number of points, annotating every single frame for large-scale sequential LiDAR point clouds is extremely laborious and expensive. For example, in a typical autonomous driving scenario, the LiDAR scanner commonly operates at a rate of 10 to 20 Hz, and each scan contains up to 10^7 points. Annotating every single point of such sequential scans to train a network would inevitably be laborious to deploy the system in practice.

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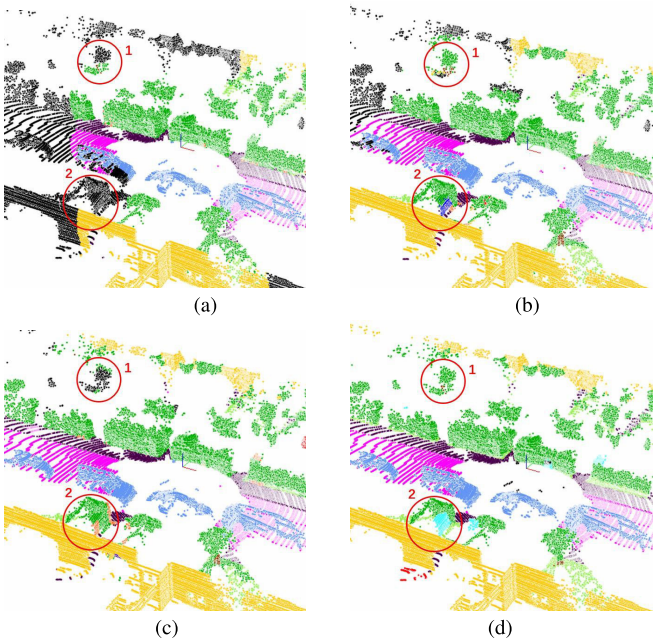


Fig. 2. Illustration of semantic segmentation results on SemanticKITTI. (a) Baseline, (b) our sparse frame-supervised method, (c) fully supervised method, and (d) ground truth.

Recently, some works introduced weakly-supervised methods to learn semantics for single point cloud scans and static scenes [11], [12], [13], [14], [15]. The weak supervision signals of these methods are primarily discrete points that are randomly chosen and annotated in whole point clouds. Fundamentally, they rely on the assumption that neighboring points share similar semantics. This is indeed effective for dense indoor static point clouds. However, for three reasons, directly extending these methods to sparse outdoor sequential point clouds is not optimal. First, the assumption of similar semantics may not be satisfied due to the sparsity of the 3-D points. Second, the persistence of object semantics over time cannot be well captured and leveraged. Third, annotating random points in every single frame is inapplicable for long sequences of dynamic point clouds. For example, the commonly-used point cloud labeler in SemanticKITTI [16] and SemanticPOSS [17] usually uses brush and polygon tools to annotate, which is vastly different from the requested random point annotation of existing methods, as illustrated in Fig. 1. It is more feasible to solely annotate a sparse number of full frames over long sequences to significantly reduce the overall annotation cost, as is done in large-scale autonomous driving datasets such as nuScenes [18], Waymo [19] and A2D2 [20].

In this article, we aim to learn the 3-D semantics of sequential LiDAR point clouds while relying only on a sparse number of point cloud frames with per-point annotations to train the networks. This is challenging, and the key challenge is how to fully exploit sparse frame annotations to effectively supervise the network. Considering that sequential point clouds inherently have the consistency of semantics over neighboring frames, we introduce a novel scene flow-based method. The key to our approach is to learn and propagate point semantics via continuous motion flows across sequential point clouds. We associated the geometric features of the successive frames and increased the semantic density of the points in a sequence of instances using sparse frame labels

Our method, named **SemanticFlow**, comprises three major components: 1) self-supervised scene flow estimation; 2) pseudolabel generation and refinement; and 3) hybrid learning of semantics. In particular, we first leverage the self-supervised method to learn rough scene flows between consecutive LiDAR point cloud frames in all training data. We then carefully designed a label propagation module to generate a set of pseudolabels for the unannotated LiDAR point cloud frames, given the sparse frames with human annotations. However, the generated pseudolabels tend to be noisy and unreliable due to motion uncertainty and environmental changes. We propose a label refinement step to enhance robustness by proposing a triplet semantic partition to avoid ambiguous semantic information in propagation and sparse generated pseudolabels. We train our semantic segmentation network in a hybrid mode with the sparse human annotations and final pseudolabels at hand. The qualitative results of our method are shown in Fig. 2.

The main contributions of this article are as follows.

1) To the best of our knowledge, we propose the first sparse frame supervised semantic segmentation network, **SemanticFlow**, on LiDAR point cloud sequences using only few fully annotated frames rather than partially annotated points in all frames.
2) We propose a pseudolabel generation and refinement method that produces many high-confidence pseudolabels for unlabeled frames.
3) We demonstrate that our approach can be integrated into multiple existing segmentation networks for point cloud sequences and achieve a significant improvement over baselines in SemanticKITTI [16], SemanticPOSS [17], and Synthia 4-D [21].

II. Related Work

A. Feature Learning on Point Clouds

Point clouds are unordered and irregular point sets in 3-D space. Traditionally handcrafted point features [22], [23] only fit specific tasks, and it is challenging to find optimal features. Recent studies convert point clouds to voxels, the regular 3-D grid [24], [25], [26], and apply CNN for feature learning. However, volumetric representations require massive memory and have inevitable information loss, which limits their performance.

Some methods process point clouds directly [27], [28], [29], [30]. As a pioneer, PointNet [1] applies feature learning to point sets and uses a symmetric function to handle different permutations of points. Subsequently, PointNet++ [2] abstracts hierarchical features and better learns local features in local point sets. However, PointNet++ [2] relies on an FPS sampling operation that requires heavy computations. RandLA-Net [31] proposes a local feature aggregation module to alleviate the information loss caused by random sampling and to improve the processing efficiency of large-scale LiDAR

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point clouds. Inspired by traditional convolution, recent methods apply convolution operations on irregular point clouds. KPConv [3] defined a set of kernel points to learn point features. SpiderCNN [4] uses the weights defined by polynomial functions to convolve points, and PointCNN [5] learns the $\mathcal{X}$-transformation matrix to process the unordered input.

Graph-based point cloud feature learning methods are also widely used [32]. Some recent methods [33], [34], [35] introduce transformer [36] into point cloud processing and perform well on multiple tasks.

B. Sequential Point Cloud Analysis

Studies on sequential point cloud analysis have recently grown in popularity. For example, PointRNN [37] leverages point-based recurrent neural networks for large-scale LiDAR autonomous driving datasets. MinkowskiNet [7] introduces 4-D spatio-temporal convolutional neural networks and 7D space-time-chroma space to process LiDAR point cloud sequences on the voxel grids. SpSequenceNet [38] proposes the cross-frame global attention module and interpolation model to process the spatio-temporal information.

MeteorNet [8] proposes direct processing of 4-D point cloud data and uses the chained flow to group local points. It can be applied to tasks such as semantic segmentation and action recognition. Furthermore, PSTNet [9] uses spatial convolution to extract local features and uses temporal convolution to process the time dimension, specializing in point-level tasks such as semantic segmentation. P4Transformer [10] employs a transformer to avoid point tracking for raw point cloud sequence modeling. Fan et al. [39] extend PSTNet and P4Transformer to propose PSTNet++ and PST-transformer [40], which can more effectively model point cloud videos. Hong et al. [41] propose a method to address the 4-D panoptic LiDAR segmentation task. HOI4D [42] presents a data collection and annotation pipeline for 4-D dynamic point cloud sequences. Moreover, flow estimation is employed to learn spatio-temporal features over sequential point cloud frames [43], [44], [45].

The above methods for semantic segmentation require full supervision, whereas our method relies on only a sparse number of annotated point cloud frames to train the networks and achieve competitive segmentation accuracy.

C. Weakly Supervised Point Cloud Semantic Segmentation

Point cloud semantic segmentation in weak supervision effectively copes with the high cost of human annotation. Using only one-tenth of the labels, Xu and Lee [11] proposed a weakly supervised semantic point cloud segmentation method and achieved comparable results with the full supervision method. Wei et al. [14] proposed a multipath region mining module for generating point-level pseudolabels. SSPC-Net [46]generates pseudolabels by passing labels over the superpoint graph. SQN [12] performs semantic segmentation with only 1‰ of labeled points under the assumption that neighboring points have similar semantic features. Zhang et al. [13] proposed learning prior knowledge from a pretext task and employing it to weakly supervised semantic segmentation on point clouds. Unal et al. [47] proposed a scribble annotation pipeline for LiDAR semantic segmentation. Yang et al. [48] proposed a weakly supervised transformer network based on multiple instance learning for indoor point clouds. Hybridcr [49] proposed a method for weakly supervised 3-D point cloud semantic segmentation using hybrid contrastive regularization. In addition, there are also some methods [50], [51], [52] that focus on weakly supervised learning for other point cloud understanding tasks such as object detection and instance segmentation.

However, the above methods are mainly designed for generating pseudolabels in a single scan point cloud and focus on the annotation cost at the point level. Shi et al. [53] used the matching algorithm to propagate pseudo-labels in the time domain. We propose a novel sparse frame supervised scheme for 3-D semantic segmentation over sequential point clouds.

III. Methodology

A. Overview

We denote a LiDAR point cloud sequence of length T as a tuple $S = (S_0, S_1, S_2, \ldots, S_T)$, in which each element is a single point cloud scan $S_t = \{p_i{}^{(t)} | i = 0, 1, 2, \ldots, n_t\}$. The point $p_i{}^{(t)}$ represents the coordinates $x_i{}^{(t)} \in R^3$. We then define the input to the multiscan semantic segmentation network as clips $C = \{C_i | i = 0, 1, 2, \ldots, T/l\}$. Clips C are divided from a point cloud sequence S with a given clip length l in the temporal order. Each clip $C_i = (S_{i\times l}, S_{i\times l+1}, \ldots, S_{(i+1)\times l-1})$ contains l neighboring frames. Recent works [7], [8], [9], [10] use sequential point clouds with full labels to perform fully supervised semantic segmentation. These labels are generally large-scale and extremely laborious to annotate. To tackle this, we define a new sparse frame supervised semantic segmentation task over sequential LiDAR point clouds using a few annotated frames.

We define the sparsity of our sparse frame supervised settings by the parameter k, where k means to annotate one frame out of every k frame. Our annotating setting requires only a small part of the annotated frames for each point cloud sequence, significantly reducing the annotation cost in practice.

This article proposes a novel SemanticFlow framework to address our sparse frame-supervised task. As shown in Fig. 3, our method comprises three parts: 1) self-supervised scene flow estimation; 2) pseudolabel generation and refinement; and 3) hybrid learning of semantics. Specifically, to generate fine pseudo-labels, we propose a novel pseudolabel propagation method based on scene flow and a label refinement scheme based on triplet semantic partition to generate one pseudolabel. In addition, we used a hybrid of generated pseudolabels and original manual labels to train a semantic segmentation model. Our framework is not restricted to any other backbone network. For computational efficiency, we selected KPConv [3], RandLA-Net [31] and P4Transformer [10] as the backbone networks for evaluation in this study.

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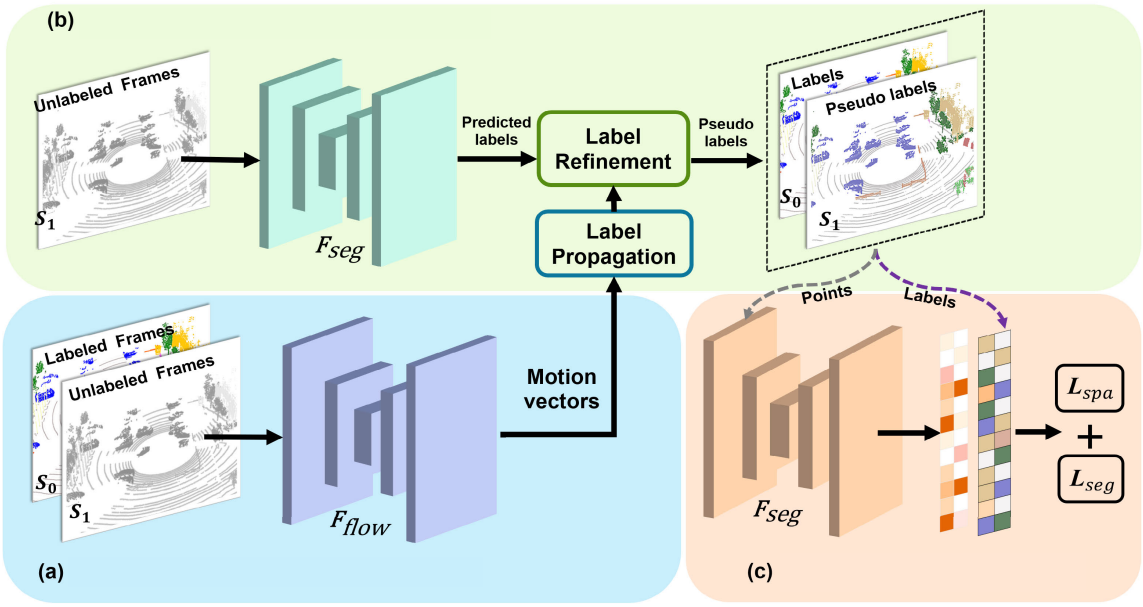


Fig. 3. Illustration of the proposed SemanticFlow method. (a) Self-supervised scene flow estimation, (b) pseudolabel generation and refinement, and (c) hybrid learning of semantics.

B. Self-Supervised Scene Flow Network

Scene flow provides a strong representation of cross-frame correlation by describing the motion state of each point. Recent studies [54], [55] have proposed several methods that can estimate scene flow without manual annotations. We adopt PointPWC [55] to perform self-supervised scene flow estimation, considering the computational complexity and the accuracy of scene flow estimation. PointPWC employs a self-supervised learning objective function to learn the scene flow in 3-D point clouds without supervision. Its loss function comprises chamfer distance, smoothness constraint, and Laplacian regularization. The network's input is a pair of 3-D point clouds S_t and S_{t+1} without annotations, and the output is a $n \times 3$ motion vector V_t of S_t.

Notably, existing scene flow networks are usually trained on the forward-facing segments of point cloud frames, ignoring motion patterns beyond the narrow field of view. Thus, they cannot directly apply to sequential point clouds covering a 360° field of view. To apply PointPWC to 360° point clouds, we first partition the whole LiDAR point cloud into four directional subpoint clouds for training and prediction and then merge them back to a complete 360° point cloud after estimating the scene flow.

C. Pseudolabel Generation and Refinement

1) Pseudolabel Propagation: Although scene flow can describe the motion state between two frames, it is still not straightforward to propagate semantic information from the previous frame to the next. As shown in Fig. 4, we designed a pseudo-label propagation module that allows the propagation of semantic information between frames to be effective. Motion vector V_t is the output of the scene flow network and describes the shift from point to point between two frames $\{S_t, S_{t+1}\}$. Thus, we can obtain S'_t, which is the estimation of frame S_t at time $t+1$ by adding the coordinates of points in frame S_t and motion vector V_t. This way, we can take semantic information to the neighbors of points in frame S_{t+1}. We then propagate the nearest semantic information from S'_t to the points in frame S_{t+1} via KNN. Finally, we generated a propagated pseudo-label ${y_i}^{\mathrm{sp}(t)}$ for each point ${p_i}^{(t+1)}$ in unlabeled frame S_{t+1}. The propagated pseudo-label ${y_i}^{\mathrm{sp}(t)}$ will be stored to $\mathrm{Label}_{\mathrm{pro}}$, defined as the set of labels from the pseudo-label propagation module.

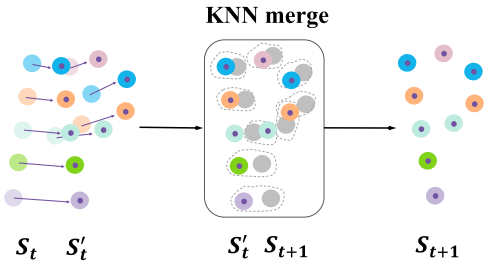


Fig. 4. Pseudolabel propagation. S_t is labeled frame. S'_t is the estimation of the S_t at time $t+1$. S_{t+1} is the unlabeled frame being propagated.

2) Bidirectional Pairs Sampling: Unidirectional propagation is a simple way to pass semantics from the first frame to the others iteratively. However, the propagated labels tended to be sparse and unreliable after several frames. We designed a bidirectional pairs sampling strategy to tackle this critical issue.

For a clip with length l, we annotate one frame in every l frames and fairly sample the pairs from both directions. We sample pairs $(S_{i\times l}, S_{i\times l+1})$, $(S_{i\times l+1}, S_{i\times l+2})\cdots$ in the first half of clip C_i, and $(S_{(i+1)\times l}, S_{(i+1)\times l-1})$, $(S_{(i+1)\times l-1}, S_{(i+1)\times l-2})\cdots$ in the second half of the clip. The labeled frame $S_{(i+1)\times l}$, which is the first frame from the next clip, is used to reverse the supervisory information. For

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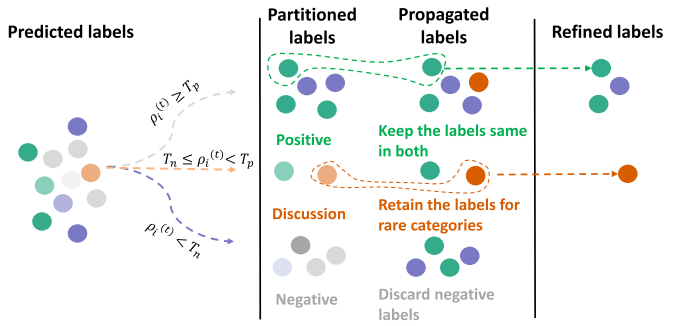


Fig. 5. Pseudolabel refinement: interpretation of triplet semantic partition. The dashed polygon represents the operation of ⊙.

example, for the first clip $C_0 = (S_0, S_1, \ldots, S_{l-1})$, we sample the pairs as follows:

$$\begin{cases} \{S_0, S_1\}, \{S_1, S_2\}, \ldots, \left\{S_{\lfloor \frac{l}{2}-1 \rfloor}, S_{\lfloor \frac{l}{2} \rfloor}\right\}, & t < \dfrac{l}{2} \\ \{S_l, S_{l-1}\}, \{S_{l-1}, S_{l-2}\}, \ldots, \left\{S_{\lceil \frac{l}{2}+1 \rceil}, S_{\lceil \frac{l}{2} \rceil}\right\}, & \dfrac{l}{2} < t. \end{cases}$$

In this way, we can obtain more supervision signals compared to unidirectional propagation.

3) Pseudolabel Refinement: The pseudolabels generated based on the scene flow inevitably contain errors and mistakes. Therefore, refining these pseudo-labels is essential and can improve the model's accuracy and training efficiency. Besides, the pseudolabel's correctness is not the only requirement. Sparsely supervised semantic segmentation requires the distribution of pseudolabel classes to be close to that of the manually annotated labels. However, classes with fewer points tend to decay more quickly during propagation and prediction, which can lead to reduced accuracy in network training due to uneven pseudolabels, even though they are correct. We first use all the labeled frames to train a prediction model F_{seg} on the backbone network and then predict these frames with pseudolabels to select labels that are more likely to be correct. The prediction result is ${y_i}^{r(t)}$ with a probability ${\rho_i}^{(t)}$. We propose a triplet semantic partition scheme to avoid ambiguous semantic information in propagation and sparse generated pseudo-labels, as is shown in Fig. 5. We designed a state mechanism and set two thresholds (T_n, T_p) to define the predicted state with the probability ${\rho_i}^{(t)}$ as follows:

$$\text{State}_{{y_i}^{r(t)}} = \begin{cases} \text{Positive}, & {\rho_i}^{(t)} \geq T_p \\ \text{Discussion}, & T_n \leq {\rho_i}^{(t)} < T_p \\ \text{Negative}, & {\rho_i}^{(t)} < T_n. \end{cases} \tag{1}$$

The default values we use of T_n and T_p are 0.25 and 0.75, respectively.

To avoid ambiguity in refinement, if $\text{state}_{{y_i}^{r(t)}}$ is negative, the result ${y_i}^{r(t)}$ will be discarded. For sparse pseudo-labels with a large and dense number of points, we set threshold T_p relatively high. If $\text{state}_{{y_i}^{r(t)}}$ is positive, the result ${y_i}^{r(t)}$ will be stored to $\text{Label}_{\text{pred}}$. To save more pseudo-labels in classes with fewer points, if $\text{state}_{{y_i}^{r(t)}}$ is discussion, ${y_i}^{r(t)}$ in classes with more points will be ignored and those in classes with fewer points will be stored to $\text{Label}_{\text{pred}}$. Finally, we use $\text{Label}_{\text{pred}}$ to refine $\text{Label}_{\text{pro}}$ to get Label_{pl} for the subsequent hybrid supervised training as follows:

$$\text{Label}_{\text{pl}} = \text{Label}_{\text{pro}} \odot \text{Label}_{\text{pred}} \tag{2}$$

where ⊙ denotes an operation that keeps the same labels and sets the elements that are not the same to void.

D. Hybrid Learning of Semantics

For hybrid learning on generated pseudo-labels and original manual labels, the loss function L_{total} includes two terms: sparsely supervised loss L_{spa} of frames with pseudo-labels and segmentation loss L_{seg} of manually labeled frames

$$L_{\text{total}} = L_{\text{seg}} + \lambda L_{\text{spa}}. \tag{3}$$

We used a softmax cross-entropy loss on the labeled points. For annotated frames, the segmentation loss is formulated as follows:

$$L_{\text{seg}} = -\frac{1}{M} \sum_{i=1}^{M} \sum_{t=1}^{n} {y_i}^{t} \log \frac{\exp({y_{\text{it}}}^{p})}{\sum_{t=1}^{n} \exp({y_{\text{it}}}^{p})} \tag{4}$$

where M denotes the number of annotated points, ${y_i}^{t}$ is the ground truth of point ${p_i}^{(t)}$ in annotated frames.

For frames with generated pseudo-labels, the loss is formulated as follows:

$$L_{\text{spa}} = -\frac{1}{M'} \sum_{i=1}^{M'} \sum_{t=1}^{n} {y_i}'^{(t)} \log \frac{\exp\left({y_{\text{it}}}^{p}\right)}{\sum_{t=1}^{n} \exp\left({y_{\text{it}}}^{p}\right)\big)} \tag{5}$$

where M' denotes the number of points with pseudo-label, ${y_i}'^{(t)}$ is the generated pseudo-label of point ${p_i}^{(t)}$ in unannotated frames.

IV. Experiments

To evaluate the performance of SemanticFlow, we conducted extensive experiments on two sequential LiDAR datasets, including SemanticKITTI [16] and SemanticPOSS [17]. In addition, our method was evaluated on a synthetic dataset–Synthia 4-D [21], which has a higher point density than the above two datasets. Furthermore, we performed ablation studies to examine the importance of the proposed main components.

A. Implementation Details

1) Dataset Setting for Scene Flow Network: We cut the original point cloud into four directional (front, back, left, and right) subpoint clouds for training and prediction to adapt the PointPWC [55] scene flow network to the above three datasets with a 360° field of view. We merged these four parts into a complete point cloud after label propagation. Furthermore, for SemanticKITTI [16] and SemanticPOSS [17], we transformed all frames into one coordinate system through poses before feeding them to the scene flow network.

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TABLE I
COMPARISON RESULTS ON OFFICIAL SEMANTICKITTI MULTIPLE SCANS TEST DATASET

Setting	Model	mIoU	mAcc	car	bicycle	motorcycle	truck	other-vehicle	person	bicyclist	motorcyclist	road	parking	sidewalk	other-ground	building	fence	vegetation	trunk	terrain	pole	traffic-sign	moving-car	moving-bicyclist	moving-person	moving-motorcyclist	moving-other-vehicle	moving-truck
Full.Sup.	TangentConv [56]	34.1	-	84.9	2.0	18.2	21.1	18.5	1.6	0.0	0.0	83.9	38.3	64	15.3	85.8	49.1	79.5	43.2	56.7	36.4	31.2	40.3	1.1	6.4	1.9	30.1	42.2
	DarkNet53Seg [16]	41.6	-	84.1	30.4	32.9	20.0	20.7	7.5	0.0	0.0	91.6	64.9	75.3	27.5	85.2	56.5	78.4	50.7	64.8	38.1	53.3	61.5	14.1	15.2	0.2	28.9	37.8
	SpSequenceNet [38]	43.1	-	88.5	24.0	26.2	29.2	22.7	6.3	0.0	0.0	90.1	57.6	73.9	27.1	91.2	66.8	84.0	66.0	65.7	50.8	48.7	53.2	41.2	26.2	36.2	2.3	0.1
	TemporalLidarSeg [57]	47.0	89.6	92.1	47.7	40.9	39.2	35.0	14.4	0.0	0.0	91.8	59.6	75.8	23.2	89.8	63.8	82.3	62.5	64.7	52.6	60.4	68.2	42.8	40.4	12.9	12.4	2.1
	Cylinder3D [58]	52.5	91.0	94.6	67.6	63.8	41.3	38.8	12.5	1.7	0.2	90.7	65.0	74.5	32.3	92.6	66.0	85.8	72.0	68.9	63.1	61.4	74.9	68.3	65.7	11.9	0.1	0.0
	KPConv [3]	51.2	89.3	93.7	44.9	47.2	42.5	38.6	21.6	0.0	0.0	86.5	58.4	70.5	26.7	90.8	64.5	84.6	70.3	66.0	57.0	53.9	69.4	67.4	67.5	47.2	4.7	5.8
Baseline	KPConv (k=5)	42.7	89.7	**88.4**	36.6	39.6	19.7	33.3	4.0	0.0	0.0	79.6	**50.9**	66.4	20.5	**81.0**	**61.9**	**73.6**	56.2	**59.5**	49.2	31.6	**63.6**	**62.5**	56.9	23.2	**8.2**	1.2
Spa. Sup.	SemanticFlow (k=5)	**45.9**	89.7	88.2	**47.5**	**46.7**	**40.8**	**39.5**	**23.3**	0.0	0.0	**80.6**	43.9	**67.1**	**24.6**	80.0	60.7	73.1	**56.5**	57.5	**50.9**	**37.5**	63.3	60.3	**58.6**	**36.0**	3.5	**6.4**

2) Self-Supervised Scene Flow Network Training: We trained the network from scratch with the Adam optimizer using the self-supervised loss released in PointPWC. The maximum number of epochs is 800. Using one Nvidia RTX 3090, we trained three models on SemanticKITTI, SemanticPOSS, and Synthia 4-D. The training time for each model is approximately 5 days, but since this is self-supervised training, only one model needs to be trained per dataset.

3) Sparse Frame Supervised Training: We employed KPConv as the main backbone network for sparse frame supervised training on three datasets and trained the network with one Nvidia RTX 3090. We conducted additional experiments on the SemanticPOSS dataset using RandLA-Net as the backbone network and used P4Transformer as the backbone network on the Synthia dataset. For the Synthia dataset, we set $k = \{10\}$ and trained the network with eight Nvidia V100. All backbone networks use our mixed total loss L_{total} and are trained to converge. The training time for this stage is consistent with the training time of the selected backbone network.

B. Evaluation of SemanticKITTI

SemanticKITTI [16] is a sequential multi-beam LiDAR dataset with dense annotation containing 28 categories, including nonmoving and moving objects. For the multiscan semantic segmentation task, 25 categories are considered to be distinguished, including moving objects and stationary ones. Sequences 00–10, containing 23 201 frames with dense annotation, are used to train the network. Sequences 11–21, containing 20 351 frames with no annotation, are used for testing by the official evaluation service.

We compared the following three kinds of methods: 1) fully supervised approaches (Full. Sup.), including the state-of-the-art methods for multiscan semantic segmentation task; 2) the baseline method implemented by KPConv without any nonessential changes; and 3) sparse frame supervised approach (Spa. Sup.), implemented by the KPConv backbone and modules proposed in Section III. For a fair comparison, we follow KPConv [3] to set clip length $l = 5$ for the baseline method and the sparse frame supervised methods. We set $k = 5$ in our experiment. Since SemanticKITTI is acquired at 10 fps, our sparse frame supervised setting parallels the frame labeling rate (2 Hz) provided by nuScenes [18] and Waymo [19].

Table I shows the results. We used the mean intersection-over-Union (mIoU) as the main evaluation metric. The scores were obtained from the official competition benchmark. When annotating one frame out of every five frames ($k = 5$), SemanticFlow gains 3.3% mIoU over baseline and is comparable to several fully supervised methods [16], [38], [56]. Significant improvements over multiple classes demonstrate the effectiveness of our label propagation in increasing semantic information density.

C. Evaluation of SemanticPOSS

SemanticPOSS [17] contains 2988 different and complicated LiDAR scans with many dynamic instances. For evaluation, we followed the experiment setting in SemanticPOSS [17] and divided the whole dataset into six parts (500 frames per part). We used data part 3 as the testing set and the others as the training set. The more balanced distribution of categories in each frame in SemanticPOSS allows for sparser annotation supervision, so we set $k = 10$.

We compared the following four kinds of methods: 1) fully supervised approaches (Full. Sup.); 2) the baseline methods implemented by KPConv and RandLA-Net without any nonessential changes; 3) sparse frame supervised approaches (Spa. Sup.), implemented by KPConv and RandLA-Net as the backbone network with modules proposed in Section III; and 4) weakly supervised approach (Weak. Sup.). We implemented the weakly supervised method proposed by Xu and Lee [11] by changing its backbone from DGCNN [32] to KPConv in the experiment for a fair comparison. Moreover, we implemented two other weakly supervised methods [13], [15] based on papers from these two approaches. We set an additional KPConv (10%) baseline trained using a dataset of 10% randomly chosen points with annotations in each frame.

We present the results in Table II. When using KPConv as the backbone network, SemanticFlow outperforms the KPConv ($k = 10$) baseline and KPConv (10%) baseline by 1.1% mIoU and 1.7% mIoU, respectively, close to the fully supervised method. When using RandLA-Net as the backbone network, SemanticFlow outperforms the RandLA-Net ($k = 10$) baseline and RandLA-Net (10%) baseline by 1.3% mIoU and 3.0% mIoU, respectively. In particular, compared with other (10%) weakly supervised methods, SemanticFlow leads up to 6.4% mIoU in results when the theoretical annotation consumption is the same.

Moreover, we illustrate a frame S_t with human annotations and the neighboring frame S_{t-1} and S_{t+1} with the generated pseudo-labels in Fig. 6(a)–(c). In addition, we compared

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TABLE II
COMPARISON RESULTS ON SEMANTICPOSS TEST DATASET

Setting	Model	mIoU	people	rider	car	traffic sign	trunk	plants	pole	fence	building	bike	road
Full.Sup.	PointNet++ [2]	20.1	20.8	0.1	8.9	21.8	4.0	51.2	3.2	6.0	42.7	0.1	62.2
	SequeezeSegV2 [59]	29.8	18.4	11.2	34.9	11.0	15.8	56.3	4.5	25.5	47.0	32.4	71.3
	KPConv [3]	55.7	79.2	22.2	65.2	30.8	75.1	26.6	37.9	83.4	57.4	54.4	80.1
	RandLA-Net [31]	53.5	69.2	26.7	77.2	24.3	27.3	73.4	30.9	51.4	82.5	43.9	81.2
Baseline	KPConv (10%)	53.5	76.5	23.4	62.9	29.4	74.1	17.5	38.1	82.1	54.6	49.8	80.1
	KPConv (k=10)	54.1	78.0	19.8	61.5	28.7	74.8	26.2	39.7	82.2	52.4	52.8	79.6
	RandLA-Net (10%)	49.4	67.1	24.6	73.4	18.6	23.5	63.8	29.9	51.7	82.1	40.6	**81.0**
	RandLA-Net (k=10)	51.1	69.3	25.0	72.5	20.2	26.1	65.9	29.5	51.8	81.3	42.8	80.3
Weak. Sup.	WS3(10%) [13]	48.8	74.4	**30.9**	42.9	26.0	70.3	7.4	31.9	78.5	43.2	52.3	79.2
	PSD(10%) [15]	50.0	75.8	28.0	57.0	25.1	68.9	9.4	35.9	78.4	42.0	50.5	79.2
	Xu (10%) [11]	54.0	78.6	19.6	69.4	**30.6**	74.0	15.0	36.8	82.3	55.4	52.5	80.2
Spa. Sup.	SemanticFlow KPConv (k=10)	**55.2**	**79.5**	19.5	67.3	28.1	**75.2**	22.6	**39.8**	**84.0**	57.7	**53.3**	79.9
	SeamnticFlow RandLA-Net (k=10)	52.4	69.8	25.7	**76.1**	20.0	26.8	**66.2**	30.3	51.5	**82.2**	43.4	80.6

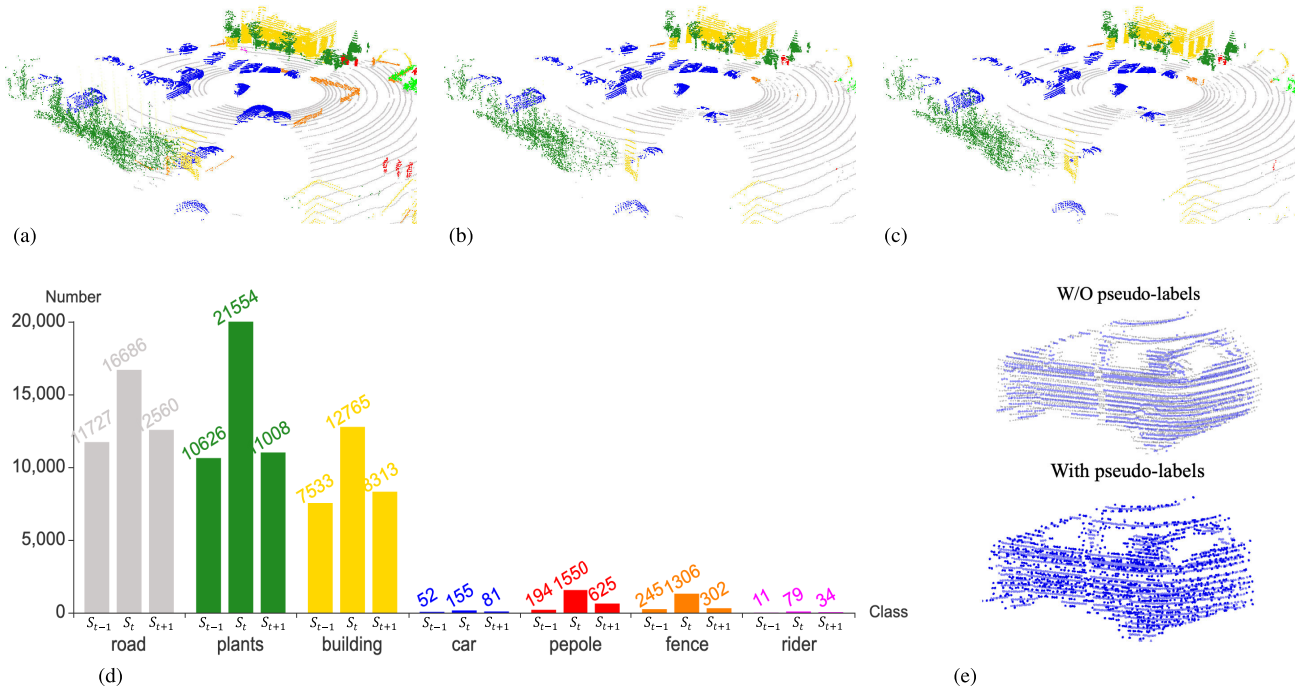


Fig. 6. Visualization and Statistics of annotated frame S_t and frames $\{S_{t-1}, S_{-2}\}$ with generated pseudo-labels in SemanticPOSS. (a) Frame S_t with manually annotated labels. (b) Frame S_{t-1} with generated pseudo-labels. (c) Frame S_{t+1} with generated pseudo-labels. (d) Statistical comparison of the number of correct labels in S_{t-1}, S_t and S_{t+1}. (e) Visualization of density comparisons after merging S_{t-1}, S_t, S_{t+1}.

the number of correct labels in these frames in Fig. 6(d), indicating that our method can propagate a large number of high-confidence pseudolabels from annotated frames S_t to unannotated frames $\{S_{t-1}, S_{t-2}\}$, both in the forward and backward propagation. Fig. 6(e) visually compares the point cloud density after merging three frames $\{S_{t-1}, S_t, S_{t+1}\}$ with and without pseudo-labels. This indicates that our approach can provide the network with denser semantic information when feeding the network with sequential frames, inducing better feature extraction and improved performance. Qualitative comparisons of semantic flow, baseline, and fully supervised methods are shown in Fig. 7.

D. Evaluation of Synthia 4-D

Synthia 4-D [7] uses the Synthia dataset [21] to create point cloud sequences. The dataset includes a collection of photo-realistic frames rendered from virtual cities with precise pixel-level semantic annotations for thirteen classes and provides six sequences of driving scenarios in nine different weather conditions. Each sequence comprised four stereo RGB-D images from four directions from the top of a moving car. The depth images were projected into the 3-D space to build the 3-D video. We follow [7] to create the point cloud sequences. We use sequences 1–4 as the training set, sequence 5 as the validation set, and sequence 6 as the testing set. The training, validation, and testing sets contain 19 888, 815, and 1866 frames, respectively. The scores in Table III were obtained from the testing set. Since this dataset provides denser points with semantic labels, we used a sparser annotation setting. Hence, we used $k = 5$ and $k = 10$ in our experiment.

We compared the following three kinds of methods: 1) fully supervised approaches (Full. Sup.), including several state-of-the-art methods on Synthia 4-D; 2) the baseline

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TABLE III
COMPARISON RESULTS ON THE SYNTHIA 4-D TEST DATASET

Setting	Model	mIoU	Bldn	Road	Sdwlk	Fence	Vegittn	Pole	Car	T. Sign	Pedstrn	Bicycl	Lane	T. Light
Full.Sup.	4D MinkNet14 [7]	77.4	90.1	98.2	73.4	87.1	99.1	97.5	94.0	79.0	92.6	0.0	50.0	68.1
	MeteorNet-l [8]	81.8	98.1	97.7	88.6	94.0	98.0	97.6	93.8	84.1	80.9	0.0	71.1	77.6
	KPConv[1]	77.6	91.6	94.5	66.7	90.6	94.4	95.3	88.9	73.9	81.0	5.9	77.6	70.5
	KPConv[2]	77.0	91.6	94.4	66.9	88.9	94.9	95.0	90.5	72.4	77.7	4.2	77.0	70.3
	P4T [10]	83.2	96.7	98.4	94.0	95.2	98.3	98.0	95.6	81.5	85.2	0.0	75.9	79.1
Baseline	KPConv(k=5)	74.4	91.2	94.3	67.8	87.2	95.3	94.3	86.9	64.7	79.9	**3.6**	74.4	53.2
	KPConv(k=10)	68.6	88.7	94.0	68.2	83.4	93.5	92.5	83.1	55.6	72.0	1.1	68.6	22.5
	P4T (k=10)	74.7	81.6	96.3	71.0	85.0	97.1	96.2	92.8	66.0	81.0	0.0	71.2	57.8
Spa. Sup.	SemanticFlow KPConv (k=5)	76.4	**93.8**	94.5	68.2	**87.5**	96.1	94.1	86.9	**73.3**	81.8	0.1	**76.4**	**64.2**
	SemanticFlow KPConv (k=10)	70.1	88.9	93.6	70.3	81.0	94.3	91.6	82.4	57.7	67.2	0.0	70.1	44.1
	SemanticFlow P4T (k=10)	**77.5**	90.7	**97.6**	**83.2**	85.6	**97.9**	**96.7**	**94.7**	71.8	**84.3**	0.0	63.9	63.5

TABLE IV
ABLATION STUDIES FOR COMPONENTS ON THE SEMANTICKITTI VALIDATION SET

Baseline	Unidir.	Bi-dir.	Refinement	k	λ	mIoU
✓				5	1	47.5
✓		✓		5	1	39.9
✓	✓		✓	5	1	48.5
✓		✓	✓	5	0.1	48.4
✓		✓	✓	5	1	**49.8**
✓		✓	✓	5	2	48.1
✓				3	1	48.6
✓		✓	✓	3	1	**50.9**
✓				10	1	36.7
✓		✓	✓	10	1	**41.1**

methods implemented by KPConv and P4Transformer without any nonessential changes; and 3) sparse frame supervised approaches (Spa. Sup.) implemented by KPConv and P4Transformer as the backbone network.

Table III shows the comparison results for Synthia 4-D. The P4T is the original model in [10] with a clip length of 3. KPConv[1] and KPConv[2] use the KPConv backbone with input clip lengths of 5 and 10, respectively. As shown in Table III, when using KPConv as the backbone network, SemanticFlow significantly improved over the baseline, improving 2.0% mIoU at $k = 5$ and 1.5% mIoU at $k = 10$ compared to the baseline result. When using P4Transformer as the backbone network, SemanticFlow still proved its effectiveness by improving 2.8% mIoU at $k = 10$. These results indicate that SemanticFlow is also effective on datasets with high-density points.

E. Ablation Study

Ablation studies were conducted to evaluate the effectiveness of each module in our framework, and the results are shown in Table IV. All ablated networks are trained on the training set, sequences {1/2/3/4/5/6/7/9/10}, of SemanticKITTI and tested on the validation split, sequence {8}, of the SemanticKITTI dataset. All networks are implemented by KPConv as the backbone network.

1) Pseudolabel Propagation and Refinement: We trained the following four semantic segmentation models to investigate the impact of each component in our designed module: 1) a baseline model trained without any label propagation and refinement; 2) a model directly trained with the raw pseudolabels output from our bidirectional label propagation module; 3) a model trained with the refined pseudolabels output from our unidirectional label propagation module; and 4) a model trained with the refined pseudolabels output from our bi-directional label propagation module. We trained all the models at $k = 5$. The experimental results are given in Table IV.

a) Importance of label refinement: As shown in Table IV, using only propagated raw pseudolabels output from our bidirectional label propagation module without refinement, the segmentation performance drops significantly by 7.6% mIoU compared with the baseline model. This is due to the abundance of noisy and false pseudolabels. These false supervision signals misguide semantics learning, consequently decreasing segmentation performance. However, by adding a label refinement scheme, our method significantly outperformed the baseline model, with a performance improvement of 2.3%, demonstrating the importance of our label refinement module.

b) Unidirectional versus bidirectional: We can simply apply unidirectional pseudolabel propagation to iteratively pass the labels forward to the next frames. However, the labels became very sparse and unreliable after several iterations. Thus, we design the bidirectional pseudolabel propagation that passes the labels both forward and backward. The performance comparison of the above two label propagation schemes is shown in Table IV. The model trained with bidirectional propagation outperforms the model trained with unidirectional propagation by 1.3% mIoU, demonstrating the effectiveness of our bidirectional label propagation design.

2) Sparsity of Annotated Frame: The interval of annotated frames is a hyperparameter in SemanticFlow. We conducted experiments on different values of k to demonstrate the generalization capability of our method. In detail, we conducted experiments on $k = 3$, $k = 5$, and $k = 10$, respectively. We compared our approach with the baseline models trained with the same value of k and the fully supervised models trained with all frames annotated. The results for SemanticPOSS, Synthia 4-D, and SemanticKITTI are given in Tables III and IV, respectively.

For the SemanticKITTI dataset, compared with the baseline method, our method showed 2.3%, 2.3%, and 4.4% mIoU improvement using $k = 3$, $k = 5$, and $k = 10$, respectively. For the SemanticPOSS dataset and Synthia 4-D dataset, our

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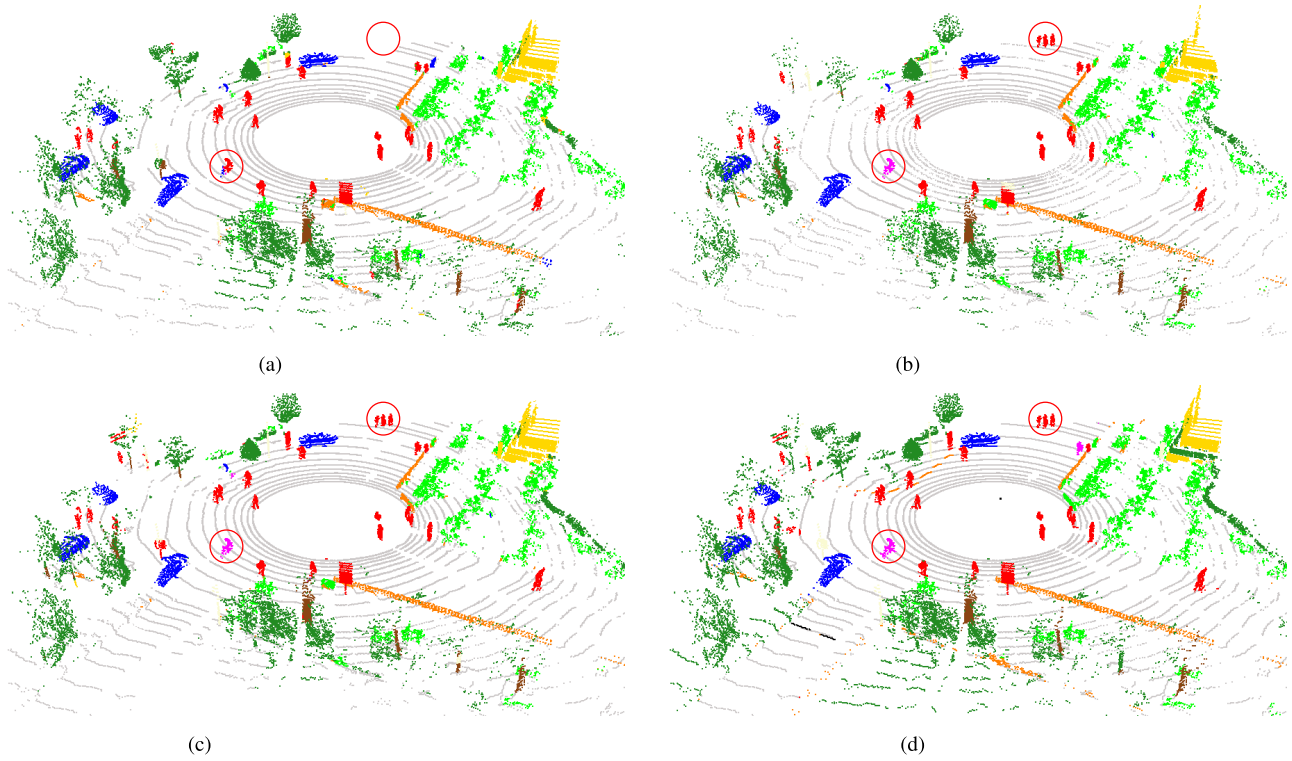


Fig. 7. Qualitative results of our method on the SemanticPoss dataset at $k = 5$. (a) Baseline, (b) SemanticFlow, (c) fully supervised method, and (d) ground truth.

method also showed a decent improvement compared with the baseline method. The results proved the generalization capability of the proposed SemanticFlow on different datasets and settings of k. However, as the sparsity increases ($k = 3$, 5 and 10), our SemanticFlow shows a decreasing trend in mIoU from 50.9% to 49.8% and 41.1%. This may occur because as the value of k increases, supervisory information requires propagation across more frames, which causes more supervisory information to be lost in the propagation.

3) Sensitivity of Hyperparameter λ: The hyperparameter λ determines the weight of the sparsely supervised loss L_{spa} in SemanticFlow. We selected three different values of λ: 0.1, 1, and 2. As shown in Table IV, the best result was achieved when λ was set to 1, with a mIoU of 49.8%. When λ was set to 0.1, the result was 48.4%, and when λ was set to 2, the result was 48.1%. We believe that when using a lower weight, i.e., $\lambda = 0.1$, the supervision effect of pseudo-labels is weakened, leading to a decrease in performance. When the λ was set to 2, there is a more significant drop in the result, which we believe is due to the inferior quality and distribution of the generated pseudo-labels compared to manually annotated labels. Moreover, since the gradient descent magnitudes of the two sub-losses L_{seg} and L_{spa} are similar, we set $\lambda = 1$ to achieve better results, which is consistent with some other weakly supervised methods [11], [46].

V. Conclusion

In this article, we propose a novel sparse frame supervised learning scheme, SemanticFlow, for 3-D semantic segmentation over sequential LiDAR point clouds. Unlike the existing weakly supervised methods, we use a sparse number of annotated point cloud frames to train the networks while achieving competitive segmentation accuracy. SemanticFlow employs self-supervised scene flow estimation to learn rough scene flows between frames. We designed a semantic label propagation module based on these flows to generate pseudolabels with sparse annotated point cloud frames. Furthermore, we propose a label refinement module using a triplet semantic partition to further improve the quality of pseudolabels. Experiments on three datasets validated the efficacy of SemanticFlow. In particular, the results of our method are compatible with some fully supervised methods.

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